ADCS Concept of Operations (ConOps)

Sebastian Benner

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1 Overview

The Attitude Determination and Control System (ADCS) is not continuously active. Instead, it operates on demand and is invoked either automatically or via commands from the ground.

The system is designed to:

- Execute specific attitude control tasks when requested
- Transition between defined operational states
- Report completion and status back to the ground

All ADCS operations follow a structured pattern consisting of an initial state, a defined task, and a shutdown operation.

2 Operational Philosophy

The ADCS operates as a task-based system rather than a continuously running control loop.

- The system is activated by either:
  - Ground command
  - Autonomous onboard trigger
- Once activated, the ADCS performs a defined task
- Upon completion, the system:
  - Sends a status message indicating completion
  - Returns to an inactive or low-power mode if no further tasks are scheduled

This approach minimizes power usage while maintaining full autonomy during task execution.

3 Operational States

The ADCS operates within a set of discrete states.

3.1 Inactive State

- ADCS control is not actively running
- Minimal power consumption
- Awaiting command from ground or autonomous trigger from C3

3.2 Active Control State

- ADCS estimation, guidance, and control loops are active
- Spacecraft is executing a defined task

3.3 Completed State

- Task objectives have been met
- Spacecraft has reached the desired attitude or condition
- Status message is transmitted to indicate completion

4 Operational Summary

All ADCS operations follow a consistent pattern:

1. Initial state (inactive or prior state)
2. Task invocation (command or autonomous trigger)
3. Task execution (control active)
4. Task completion
5. Status reporting
6. Transition to end state (inactive or standby)

This structure ensures predictable behavior, low power consumption, and clear communication of system status.

4.1 Nominal Operational Sequence

A typical operational sequence follows the order below:

1. Ground command is received specifying:
   - Mission type
   - Duration
   - Whether data should be offloaded after completion
2. The ADCS is initialized:
   - State estimation is initialized
   - If necessary, the spacecraft performs a slow rotation about the +z axis until the star tracker is no longer occluded
3. The spacecraft executes the commanded mission for the specified duration
4. Upon completion, the system evaluates upcoming ground station overpasses
5. If the next overpass is sufficiently far in the future:
   - The spacecraft enters a low-power mode
   - ADCS control is shut down
6. Prior to the overpass window:
   - The ADCS is reinitialized if previously shut down
   - The spacecraft transitions to target tracking mode

7. Data is offloaded during the overpass
8. After reaction wheel shutdown, a detumble operation is performed:
   - Executed at a lower update rate
   - Ensures the spacecraft is in an acceptable state for the next mission
   - Results in a small residual spin about the +z axis
9. The system sends a final status message and transitions to the inactive state

5 Detumble Operation

5.1 Trigger Conditions

Detumble is initiated automatically following deployment.

The trigger condition is:

- A predefined time delay after deployment

This ensures that the spacecraft begins stabilizing without requiring ground intervention.

5.2 Execution

- Magnetorquer-based control is used to reduce angular velocity
- The system monitors angular rate until it falls below a defined threshold

5.3 Completion

- Once angular rates are sufficiently reduced:
  - The detumble task is considered complete
  - A status message is sent to the ground
- The spacecraft transitions to a stable state and awaits further commands

6 Mission Execution

6.1 Commanded Operation

Mission execution is initiated via ground command.

The command specifies:

- Mission type (e.g. nadir pointing, ram-facing orientation)
- Duration of execution
- Whether data should be downlinked after completion

6.2 Execution Behavior

Once commanded:

- The ADCS transitions from the inactive state to the active control state
- The spacecraft performs the requested mission for the specified duration

6.3 Attitude Modes

- **Nadir Pointing:** Spacecraft +z axis aligned with Earth center
- **Ram-Facing:** Spacecraft +x axis aligned with velocity vector

For circular or near-circular orbits, these modes are nearly equivalent. In elliptical orbits, the difference becomes more pronounced due to variation in the velocity vector.

6.4 Completion

- After the specified duration:
  - The task is terminated
  - A status message is transmitted
- The spacecraft transitions to the completed or inactive state

7 Autonomous Downlink Operation

7.1 Overview

If data downlink is requested, the spacecraft autonomously schedules communication following mission completion.

7.2 Overpass Prediction

- The spacecraft predicts upcoming ground station overpasses
- The next suitable overpass window is selected

7.3 Pre-Overpass Behavior

- The spacecraft remains in a low-power or standby state
- ADCS is not actively performing high-rate control during this period

7.4 Communication Execution

- Shortly before the overpass window:
  - ADCS is activated
  - The spacecraft transitions to target tracking mode
- The spacecraft points toward the ground station
- Data transmission begins once the ground station is within range

7.5 Completion

- After the overpass:
  - Transmission stops
  - A status message is generated
- The spacecraft returns to an inactive or standby state